Yes. I think Alex has identified a **much better route for the purpose of ASSIGN**.
The important shift is this: we probably do **not** need to invent a new network-growth model. We need a defensible, minimal **constrained bipartite assignment mechanism**. The literature already gives us the structural skeleton, and Quayle gives us the behavioral bias to put inside it.
The clean story becomes:

> ***Start from a bipartite configuration model that controls the degree sequence/capacities, then bias otherwise allowable individual–team assignments using Quayle-style similarity attraction.***

That is substantially easier to motivate than "we invented a new formation model."

Why the pieces fit

A bipartite configuration model has exactly the structural property Alex was reaching for. In a bipartite graph with individuals on one side and teams on the other, we specify degree requirements on both layers. For our snapshot:

$$k_i = 1$$

for each individual—one affiliation—and

$$k_j = C_j$$

for each team if all roster positions are filled.
With uniform basketball capacity,

$$k_j = C = 15.$$

Therefore total stubs balance:

$$\sum_i k_i = \sum_j C_j.$$

Ordinary bipartite configuration matching would then connect individual stubs to team stubs without caring about performance similarity. **That's precisely the place where Quayle can enter.**
Instead of making the next allowable match uniformly, give individual i's possible connection to team j the Quayle-inspired weight

$$\boxed{w_{ij} = e^{-\rho|A_i - \mu_j|}}$$

and sample among teams with remaining stubs:

$$\boxed{P(i \to j) = \frac{R_j\, e^{-\rho|A_i - \mu_j|}}{\sum_l R_l\, e^{-\rho|A_i - \mu_l|}}}$$

where

$$R_j = C_j - n_j$$

is team j's **remaining capacity**.
That equation is worth staring at, because I think it may be the whole ASSIGN mechanism.
At

$$\rho = 0,$$

it becomes

$$P(i \to j) = \frac{R_j}{\sum_l R_l},$$

which is exactly what you would expect from random matching of an individual stub to the remaining **team stubs** in a configuration-style construction.
As ρ increases,

$$e^{-\rho|A_i - \mu_j|}$$

biases those structurally available stubs toward teams whose centroids resemble i.
And when

$$R_j = 0,$$

the team's probability automatically becomes zero.

No special "remove the team from rotation" code is conceptually necessary. Capacity is already built into the configuration mechanism. That is elegant.

One important refinement to our earlier thinking

Notice the R_j term.

Earlier, we had been considering:

$$P(i \to j) = \frac{e^{-\rho d_{ij}}}{\sum_{l \in \mathcal{F}} e^{-\rho d_{il}}}.$$

That makes every *team* with space one alternative regardless of whether it has 1 opening or 10 openings.

A configuration-model interpretation says something subtly different and, I think, more principled: **the objects being matched are stubs**. A team with 10 remaining positions has 10 available team stubs; one with 1 remaining position has one.

Hence:

$$\boxed{\text{remaining capacity} \times \text{Quayle similarity}}$$

becomes the attachment weight.

That isn't an arbitrary new structural-prestige mechanism like Quayle's $f(k_j)$. It's simply the combinatorics of the configuration model.

And the literature supports the foundation. Saracco et al.'s Bipartite Configuration Model (BiCM) explicitly extends configuration-model reasoning to binary undirected bipartite networks by constraining degree sequences on both layers. Later maximum-entropy work develops this framework extensively. There is also a separate literature on **bipartite ERGMs with homophily**, which is useful confirmation that bipartite degree structure and attribute-dependent edge preference can coexist mathematically, although I would *not* jump to a full ERGM unless we need it—the complexity would defeat Alex's objective.

Alex's projection insight is good, with one qualification

Yes, Quayle's one-mode network can help us conceptually think about what we see after projecting affiliations onto individuals: members of the same team become connected/co-members in the projection.

But I would **not claim that Quayle's model literally is the projection of our bipartite process**. That's stronger than we've established. Projection induces cliques/co-membership structure and degree dependencies that Quayle's growth process does not explicitly generate.

The safer—and more interesting—statement is:

> *Quayle provides the **micro-level similarity mechanism** we want the affiliation process to reproduce, while the bipartite configuration model supplies the **structural constraints** that Quayle lacks.*

That's enough. We don't need projection equivalence.

The model may now be much simpler than what we were building

For a given formation period t, freeze team centroids:

$$\mu_{j,t-1}.$$

Each entrant i has ability:

$$A_{i,t-1}$$

or an empirically informed entrant draw.

Each team begins with remaining roster capacity:

$$R_{j,t} = C_j - n^{\text{inc}}_{j,t}.$$

Randomly order the players requiring assignment.

Then sequentially draw:

$$\boxed{P(i \to j \mid t, s) = \frac{R^{(s)}_{j,t} e^{-\rho|A_i - \mu_{j,t-1}|}}{\sum_l R^{(s)}_{l,t} e^{-\rho|A_i - \mu_{l,t-1}|}}}$$

and after assigning i to j,

$$R^{(s+1)}_{j,t} = R^{(s)}_{j,t} - 1.$$

Continue until all individual stubs and team stubs are exhausted.

That's it.

We still have the dynamic feedback afterward:

$$\mu_{j,t-1} \to \text{configuration-style biased matching} \to T_{j,t} \to \mu_{j,t}.$$

But we no longer need to pretend we're designing an entire network-growth theory.

And the interpretation of ρ stays beautiful

At

$$\rho = 0$$

we recover the random bipartite configuration-style assignment baseline.
For

$$\rho > 0$$

we increasingly bias stub matching by similarity.
For very large ρ, assignments become strongly assortative **subject to the degree/capacity constraints**.

That last clause is scientifically interesting:

$$\boxed{\text{homophilic preference} \neq \text{perfect assortative realization}}$$

because capacity can prevent everyone from receiving their most similar affiliation.

That is precisely the distinction we liked in Quayle—**generative preference versus realized assortativity**—now transplanted into the structure we actually need.

And we absolutely should **not** use Quayle's analytic $\alpha \leftrightarrow r$ formula. Our capacity constraints and bipartite matching change the mapping. We can simply simulate the relationship

$$\rho \longrightarrow \text{realized sorting}$$

and report it.

The literature path I would take now

I would **stop hunting for a magical paper that already contains our exact model**. We don't need one. I think the defensible lineage is:

1. **Configuration model / bipartite configuration model** for the constrained bipartite matching skeleton.
2. **Quayle et al. (2006)** for the exponential similarity/homophily bias and the crucial preference-versus-assortativity distinction.
3. **Bipartite ERGM/homophily literature** as supporting precedent that attribute-biased formation in bipartite networks is well established—not necessarily as our implementation.

The most directly useful BiCM anchor I found is Saracco et al. (2015), which explicitly develops the bipartite extension of the configuration model with degree constraints. A recent review of bipartite pattern detection also distinguishes hard degree-preserving configuration models from softer maximum-entropy variants. And Bomiriya et al. specifically study homophily in exponential-family random graph models for bipartite networks, which is conceptually adjacent to what we're doing.

Saracco, F., Di Clemente, R., Gabrielli, A., & Squartini, T. (2015). Randomizing bipartite networks: the case of the World Trade Web. Scientific Reports, 5, 10595. https://www.nature.com/articles/srep10595

Straka, M. J., Caldarelli, G., & Saracco, F. (2017). Grand canonical validation of the bipartite international trade network. https://arxiv.org/abs/1703.04090

Saracco, F., et al. (2018). From ecology to finance (and back?): A review on entropy-based null models for the analysis of bipartite networks. Journal of Statistical Physics. https://link.springer.com/article/10.1007/s10955-018-2039-4

Bomiriya, R. P., Kuvelkar, A. R., Hunter, D. R., et al. (2023). Modeling homophily in exponential-family random graph models for bipartite networks. https://arxiv.org/abs/2312.05673

Quayle, A. P., Siddiqui, A. S., & Jones, S. J. M. (2006). Modeling network growth with assortative mixing. European Physical Journal B, 50, 617–630. https://link.springer.com/article/10.1140/epjb/e2006-00170-5

The next thing I would do is **not code yet**. I would spend a short, tightly bounded literature pass verifying whether the exact weighted-stub rule

$$R_j e^{-\rho d_{ij}}$$

already has a recognized name/model lineage, and whether hard degree preservation introduces any sampling subtlety we need to acknowledge. If it checks out, I think we may have turned what looked like a weekend rabbit hole into a very small ASSIGN module with an extremely clean methodological pedigree.